PO12Q: Introduction to Quantitative Political Analysis II

Week 10 - Worksheet Solutions

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# 1 | Exercises with the app

1. Sampling variability and the true parameter
   a) The three values of $\hat{\beta}_1$ will differ from each other and from the true value of 1. Each simulation draws a fresh random sample, so the estimate varies due to sampling variability – the error term $\varepsilon$ is different each time.
   b) With $n$ = 1000 the estimates cluster much more tightly around $\beta_1$ = 1. Larger samples give OLS more information, reducing the standard error of the estimator.

   **Takeaway:** OLS estimates are random variables. They scatter around the true $\beta_1$, but larger samples bring them closer. This is the difference between an estimator (the procedure) and an estimate (one realisation).

2. What heteroskedasticity does – and doesn't – break
   a) $\hat{\beta}_1$ does not shift systematically; it stays centred on the true value. Heteroskedasticity does not bias OLS coefficients – they remain unbiased.
   b) When the Breusch-Pagan test rejects, the OLS standard errors are unreliable because they assume constant variance. The HC3 (robust) standard errors are valid because they do not rely on the homoskedasticity assumption. In practice, the HC3 SEs are typically larger, making inference more conservative but more honest.

   **Takeaway:**
   Heteroskedasticity breaks the formula for standard errors, not the coefficients themselves. Conventional OLS SEs assume constant variance, so when that fails, confidence intervals and p-values become unreliable. Robust (HC3) SEs fix this.

3. The cost of ignoring nonlinearity
   a) The linear fit cuts through the curve, systematically over-predicting $y$ in the middle range of $x$ and under-predicting at the extremes (or vice versa, depending on the sign of $\beta_2$). The residuals show a clear U-shaped or inverted-U pattern.
   b) The F-test p-value drops below conventional thresholds (0.05, 0.01) once $\beta_2$ is large enough for the curvature to be detectable given the sample size. Fitting a straight line to curved data produces biased predictions – especially at the tails of $x$, where extrapolation errors compound.

   **Takeaway:**
   Unlike heteroskedasticity, nonlinearity biases OLS estimates. The slope $\hat{\beta}_1$ no longer estimates a meaningful quantity because the linear model is the wrong functional form. Residuals carry systematic patterns, and predictions at the extremes of $x$ can be badly wrong.

4. Why collinearity makes you uncertain, not wrong
   a) As $\rho$ rises toward 1, VIF increases sharply (e.g. exceeding 10 or even 100), and standard errors for both $x_1$ and $x_2$ inflate correspondingly. The estimates become very imprecise.
   b) The estimate of $\beta_1$ changes when $x_2$ is dropped because $x_1$ absorbs the effect of the omitted correlated variable. If $x_2$ is a genuine confounder, only the model including both regressors gives a coefficient with a ceteris paribus interpretation – but the price is large standard errors.
   c) Yes. At high $\rho$, $\hat{\beta}_1$ and $\hat{\beta}_2$ swing wildly from one simulation to the next, even though on average they remain centred on their true values.

> **Takeaway:**
> Collinearity inflates variance but does not introduce bias – OLS is still unbiased, just imprecise. The real danger is instability: estimates swing wildly across samples because the data cannot distinguish each variable's separate contribution. This is a problem of information, not of the estimator.

5. Connecting violations to the Gauss-Markov conditions

| Assumption | Heteroskedasticity | Nonlinearity | Collinearity |
|---|---|---|---|
| $E(\varepsilon \mid X) = 0$ | Satisfied: errors are still mean-zero at every $x$; only their spread changes. | Violated: the omitted curvature is absorbed into $\varepsilon$, so $E(\varepsilon \mid X) \neq 0$. | Satisfied: the DGP is linear in both $x_1$ and $x_2$, so the conditional mean is correctly specified. |
| $\mathrm{Var}(\varepsilon \mid X) = \sigma^2$ | Violated: the error variance changes with $x$, so it is no longer constant. | Satisfied: once the correct functional form is specified, the errors have constant variance. | Satisfied: the error variance does not depend on $x_1$ or $x_2$. |
| No perfect multicollinearity | Satisfied: there is only one regressor. | Satisfied: there is only one regressor. | Approached: $x_1$ and $x_2$ are near-perfectly correlated, so the $X'X$ matrix is close to singular. |
| OLS unbiased? | Yes: $E(\varepsilon \mid X) = 0$ still holds, so OLS remains unbiased. | No: the misspecified model omits $x^2$, which is correlated with $x$, biasing $\hat{\beta}_1$. | Yes: all assumptions for unbiasedness hold; the problem is precision, not bias. |
| OLS SEs valid? | No: the standard SE formula assumes $\sigma^2 I$, which fails here; use robust (HC) SEs. | No: the model is wrong, so both coefficients and SEs are unreliable; fix the functional form first. | Yes in principle, but they are very large because near-collinearity inflates $\mathrm{Var}(\hat{\beta})$ via VIF. |

Table 1: How each violation maps to the Gauss–Markov conditions

> **Takeaway:**
> Different violations break different properties. Knowing which assumption fails tells you what goes wrong – and therefore which remedy to apply (robust SEs for heteroskedasticity, a nonlinear specification for functional form misspecification, or dropping/combining regressors for collinearity).

## 2 | Testing the Classical Linear Assumptions

see RScript.